Demonstration of Proposed Features

Date	
Team ID	
Project Name	OptiCrop - Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Crop Recommendation Engine	Predict the most suitable crop based on soil nutrients (N, P, K, pH) and environmental parameters (temperature, humidity, rainfall). Display predicted crop name, confidence score, and detailed crop information.	Implemented	Yes	
2	Suitability Assessment	Evaluate how suitable the given conditions are for crop cultivation. Return suitability status (Optimal/Moderate/Poor), productivity level, per-factor environmental compatibility, soil analysis, and tailored recommendations.	Implemented	Yes	
3	Research Dashboard	Display comprehensive dataset analytics including summary cards, model comparison table with best-model highlighting, Chart.js bar chart, dataset statistics, environmental summary, feature correlations,	Implemented	Yes	

		K-Means cluster analysis, and 10 chart images.			
4	Multi-Model Training & Comparison	Train and evaluate 6 ML classifiers (Random Forest, SVM, Decision Tree, Logistic Regression, KNN, Naive Bayes) with cross-validation. Save metrics and identify the best-performing model.	Implemented	Yes	
5	K-Means Exploratory Clustering	Apply unsupervised K-Means clustering on PCA-reduced data to discover natural groupings in the dataset. Visualize clusters alongside actual crop labels for comparison.	Implemented	Yes	
6	Chart Image Generation	Automatically generate 10 publication-quality chart images covering crop distribution, nutrient analysis, environmental parameter distributions, correlation heatmap, boxplots, pairplot, and K-Means visualization.	Implemented	Yes	

Feature Implementation Summary:

Total Features Proposed	8
Total Features Implemented	8
Total Features Demonstrated	8
Overall Implementation Rate (%)	100%